

Corporate Financing and Investment Decisions When Firms Have Information That Investors Do Not Have

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ABSTRACT

Myers and Majluf ask what happens to a firm's issue–invest decision when managers know more about the value of the firm's assets and growth opportunities than the outside investors who must fund a new project. Building a three-date rational–expectations model in which a stock issue is the only channel for new information to reach the market, they show that an equity issue is always read as bad news, because managers only issue when the market would otherwise overvalue the firm's existing assets. Firms therefore sometimes decline positive–net–present–value projects rather than sell undervalued shares, a real, non–Modigliani–Miller cost of external equity that internal funds and riskless debt do not carry. The model yields an ordered financing preference—retained earnings, then debt, then equity—that became known as the pecking order, together with a rationale for holding financial slack and a prediction, later confirmed empirically, that share prices fall on the announcement of a seasoned equity offering. The analysis rests on a specific assumption about managerial loyalty to existing shareholders and a stylized information structure, both of which this review examines critically alongside the model's subsequent empirical record.

1. Introduction

By 1984, the Modigliani–Miller framework had trained a generation of financial economists to treat the choice between debt, equity and retained earnings as a matter of taxes and distress costs layered onto an otherwise irrelevant decision. PAPER Myers and Majluf's contribution is to show that irrelevance breaks down for a reason unrelated to taxes or bankruptcy: the mere act of selling equity to outsiders who know less than management about firm value forces a transfer of wealth between old and new shareholders, and that transfer alone can be large enough to make managers walk away from a genuinely valuable project.

The paper's central discovery is that this friction is not merely a pricing inefficiency but an equilibrium implication of rational behaviour on both sides of the market. Investors, aware that managers will only issue stock when doing so benefits existing shareholders, mark down any issuing firm's share price on the news of an issue itself—independent of what the issue proceeds will be used for. Managers, anticipating this markdown, sometimes prefer to forgo the investment rather than trigger it. The result is a self-reinforcing equilibrium in which the correlation between an equity announcement and a lower valuation is not a market failure but the market working exactly as rational-expectations theory says it should.

What was missing from the literature before this paper was a fully worked equilibrium model connecting three things that had previously been treated separately: the lemons problem of adverse selection (*Akerlof, 1970*), the growing signalling literature on capital structure (*Ross, 1977*; *Leland and Pyle, 1977*), and the empirical observation, then only beginning to accumulate, that share prices fall when firms announce new equity. Myers and Majluf supply the missing equilibrium: a closed-form partition of the space of possible firm values into an "issue" region and a "do not issue" region, a zero-profit break-even price for investors conditional on that partition, and a set of comparative statics linking financial slack, project size and asset-value uncertainty to the expected loss from underinvestment.

The finding matters for two audiences. For corporate-finance theorists, it establishes that the seemingly behavioural aversion managers display toward equity issuance—"we do not want to be undervalued"—survives formalisation as a fully rational equilibrium outcome, and it produces a specific ranking of financing sources, later termed the pecking order (*Myers, 1984*), that is falsifiable against firm-level data. For empiricists, it supplies the first rigorous theoretical account of why announcement-period returns to seasoned equity offerings are reliably negative while debt announcements are not, a pattern documented contemporaneously by *Dann and Mikkelson (1982)*, *Korwar (1982)* and *Asquith and Mullins (1983, 1986)*. This review reconstructs the model in full, reproduces its central equations and numerical experiments, and then subjects both the theory and its subsequent empirical fate to critical scrutiny.

2. Research Problem and Hypothesis

2.1 Research Question

The paper asks: when a firm's managers possess private information about the value of both its existing assets and a new investment opportunity, and must issue common stock to outside investors to finance that opportunity, does a fully rational equilibrium exist in which the firm sometimes declines a positive-net-present-value project rather than issue equity—and if so, what determines the size of that underinvestment and the ranking of financing instruments a value-maximising firm will choose?

2.2 Motivation

The question matters because it reopens, on purely informational grounds, a debate that the Modigliani–Miller theorems had supposedly closed. If financing choice can, on its own, destroy real investment value even absent taxes, transaction costs or bankruptcy risk, then capital structure is not just a slicing of a fixed pie but an input into the size of the pie itself. A reader of this review should care about this paper because its answer reorganises how corporate liquidity, dividend policy, merger motives and the choice between debt and equity are subsequently taught and tested: financial slack stops being idle cash and becomes an option to invest without incurring an information-driven discount.

2.3 Existing Literature

Three literatures converge in this paper without, prior to it, having been combined. First, *Akerlof (1970)* had shown that adverse selection about the quality of a single good can shut markets down entirely. Second, a signalling literature—*Ross (1977)*, *Leland and Pyle (1977)*, *Bhattacharya (1979)*—had modelled how a firm's capital-structure or dividend choice can convey private information to the market, typically in stylised two-type settings. Third, *Miller and Rock (1982; published 1985)* had modelled dividend policy under asymmetric information about current cash flow, predicting that unexpectedly large external financing is bad news. What none of these strands provided was an equilibrium model in which a firm holds a claim on two assets simultaneously—assets already in place and a new project—and must sell a package containing both to finance the second. PAPER Myers and Majluf note explicitly that their problem "is similar to the one addressed by Akerlof," but that the joint claim on two assets, one of which the firm forfeits entirely if it does not sell the package, gives their model a structure and a set of predictions that a single-good lemons model cannot produce.

2.4 Research Gap

The gap was a formal, closed-form equilibrium linking three ingredients that mattered to practitioners and had been observed anecdotally—managerial reluctance to issue stock, the empirical fact of negative announcement returns to equity offerings, and firms' apparent

preference for internal funds and debt—without any of the existing signalling or lemons models tying all three together in one internally consistent setting with a rational, price-taking market on the other side.

2.5 Contribution

PAPER The authors claim to supply (i) a fully solved equilibrium model of the issue–invest decision, (ii) a proof that debt dominates equity whenever asymmetric information concerns firm value rather than risk, (iii) numerical estimates of the resulting loss in firm value, and (iv) a set of extensions to mergers, dividends and dynamic financing. **CRITIQUE** The claim is largely made good within the paper's own assumptions: the equilibrium is genuinely closed-form for the issue–invest boundary and for the debt-versus-equity ranking, and the numerical experiments (Section 5 below) do quantify the loss under specific distributional assumptions. Where the contribution is thinner than advertised is in claim (iv): several of the extensions—temporary information asymmetry, mergers—are argued informally rather than solved as fully as the core model, a point the authors themselves flag ("we will not here pursue analysis of the optimal issue strategy in this dynamic setting").

3. Theoretical Framework and Mechanism

3.1 The Three-Date Structure

The model has three dates. At $t = -1$, managers and the market share the same information about the firm. At $t = 0$, managers privately learn the realised value of assets already in place and of a new investment opportunity; the market does not learn this until $t = +1$. The issue–invest decision must be taken at $t = 0$, precisely when the information asymmetry is at its widest.

A, a	The random variable representing the value of assets already in place, and its realisation known to managers at $t = 0$ (negative values ruled out by limited liability).
B, b	The random variable representing the net present value of the new investment opportunity, and its realisation known to managers at $t = 0$ (truncated at zero, since a negative-NPV project is simply not undertaken).
S	Financial slack: cash and marketable securities (or riskless borrowing capacity) available to fund the project without a stock issue, fixed and observed by both sides.
I	The investment required by the new project.
E	The amount of new equity required, $E = I - S$, when $S < I$.
P'	The market value of the old shareholders' stake at $t = 0$ conditional on the firm issuing stock and investing.
P	The market value of the old shareholders' stake at $t = 0$ conditional on the firm not issuing.

Management is assumed to act in the interest of the "old" shareholders—those who owned the firm's shares before any new issue—and, in the core version of the model, those shareholders are assumed passive: they neither buy the new issue nor rebalance their portfolios in response to the firm's decision. This assumption, examined critically in Section 7 below, is what makes financing non-neutral; Section 3.4 shows what happens when it is relaxed.

3.2 The Issue–Invest Condition

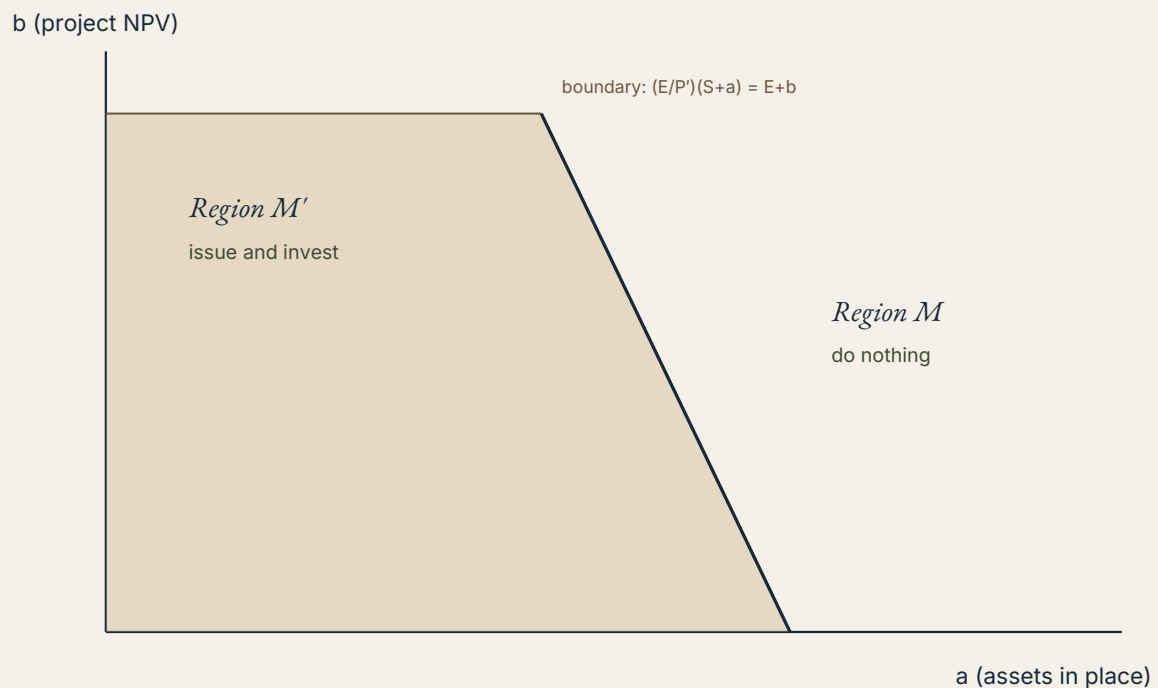
Old shareholders are better off if the firm issues and invests whenever the value they retain after the issue exceeds what they would keep by doing nothing, $S + a$. Since issuing dilutes their claim to a fraction $P/(P + E)$ of a total post-investment value of $E + S + a + b$, the condition for issuing is:

$$(EP')(S + a) < E + b \quad (1)$$

The left-hand side is the share of existing assets and slack that old shareholders surrender to new investors; the right-hand side is the increment to firm value—the new project—that old shareholders capture by going ahead. The firm issues only when what it keeps of the upside (b) exceeds what it gives away of the downside (its share of $S + a$). Because P' itself depends on which realisations of (a, b) trigger an issue, and the issue decision depends on P' , the model closes with a rational-expectations zero-profit condition on the price investors pay:

$$P' = S + A(M') + B(M') \quad (2)$$

where $A(M')$ and $B(M')$ are the expectations of assets-in-place and project NPV conditional on the realisation falling in the issuing region M' —that is, conditional only on what the decision to issue itself reveals. Equations (1) and (2) are solved jointly: the boundary $(EP')(S + a) = E + b$ partitions the joint support of (A, B) into a region M' ("issue and invest") and its complement M ("do nothing"), and P' must be the price that makes investors break even given exactly that partition.

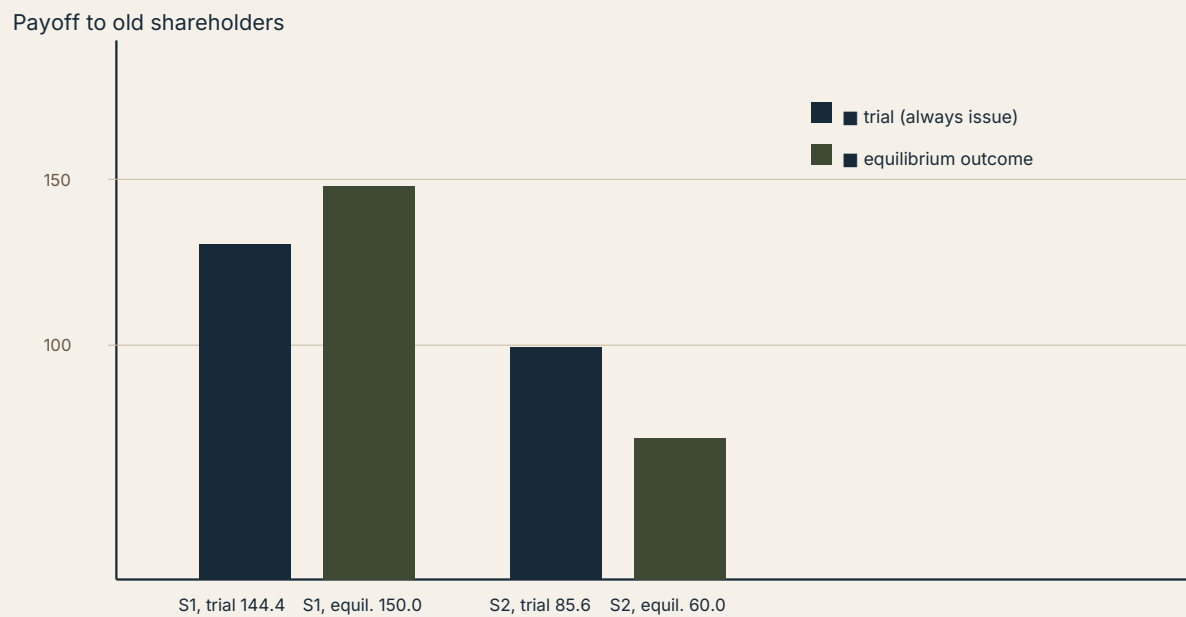
Figure 1. The issue–invest partition of the (a, b) plane

Schematic reconstruction of the paper's Figure 1 (not to scale; illustrates the qualitative partition described in the text, not the authors' original coordinates). Source: author's reconstruction from Myers and Majluf (1984), Section 2.

The region M' sits toward low a and high b : the firm is most willing to issue precisely when its existing assets are worth little (so it has little to lose by selling a stake in them) and its new project is worth a great deal (so the gain from proceeding outweighs the dilution). Conversely, in region M , high a means the firm has a great deal to lose from selling undervalued shares, and it declines the project even when $b > 0$.

3.3 The Numerical Example and the Unravelling Mechanism

The paper's two-state example makes the mechanism concrete. With two equally likely states, assets-in-place worth $a = 150$ or $a = 50$, and project NPV $b = 20$ or $b = 10$ respectively, a trial equilibrium in which the firm always issues and invests gives old shareholders 144.42 in state 1 and 85.58 in state 2. But state 1's old shareholders would be better off simply keeping the 150 and forgoing the project than accepting 144.42 for issuing. Once the firm follows that incentive and withholds the issue in state 1, the issue decision itself reveals state 2, so the price conditional on issuing collapses to $P' = 60$, and the firm passes up a project worth 20 in state 1 purely because issuing would out it as the low-value type.

Figure 2. Payoffs to old shareholders: trial versus equilibrium outcomes in the two-state example

Values reconstructed from the numerical example in Myers and Majluf (1984), Section 1 (author-generated chart, not from the original paper). State 1 old-shareholder value rises from 144.4 to 150.0 by withholding the issue; state 2 value falls from 85.6 to 60.0 once the withholding in state 1 reveals state 2 whenever an issue is observed. The ex ante average value to old shareholders falls from 115 to 105, a loss of 10, entirely attributable to the forgone state-1 project.

3.4 Why the Result Depends on Passive Shareholders

Table 2 of the paper shows that the underinvestment result is not an artefact of where cash happens to sit. If old shareholders hold $S = I$ in their own accounts and must sell part of their stock to recoup $(1-\alpha)I$ after the firm invests internally, the equilibrium condition is identical to the case in which the firm raises the same amount externally. **INFERENCE** The active ingredient is not the location of the cash but whether shareholders' portfolio responses to the firm's decision are constrained. If old shareholders can be compelled to buy and hold a pro-rata share of any new issue ($\alpha = 1$ in the paper's notation), the model collapses back to a Modigliani–Miller world: the firm always invests when $b > 0$, and the ex ante loss L disappears. The authors are explicit that this is why "passive" old shareholders are the load-bearing assumption of the entire analysis, not a simplifying convenience.

3.5 Debt versus Equity: the Pecking Order Result

Extending the model to a choice between risky debt and equity, the firm issues and invests only if the project's NPV covers the capital gain the new security holders would realise once the truth is known:

$$S + a < S + a + b - \Delta E \quad (3)$$

where ΔE is the ex post gain (or loss) to new shareholders once a and b are revealed, priced at equilibrium to average to zero. Because option-pricing logic implies ΔD and ΔE share the same sign but $|\Delta D| < |\Delta E|$ for any risky debt claim junior only to default risk, a firm willing to issue equity is always willing to issue debt, but not the reverse: debt is issued in states where equity is not, strictly enlarging the region in which the firm invests. **PAPER** The authors go further and prove that, given a discretionary choice at $t=0$, no equilibrium price exists at which the firm would ever choose to issue equity over riskless or default-risky debt when the asymmetry concerns firm value rather than variance: any such price would require new equity holders to accept a sure expected loss. This yields the paper's best-known normative statement—retained earnings first, then debt, then equity as a last resort—subsequently named the pecking order by *Myers (1984)*.

4. Data and Empirical Strategy

This is a theoretical paper; it has no firm-level data set of its own. Two distinct forms of quantitative discipline are nonetheless present and worth separating clearly. **PAPER** First, Section 2 of the paper reports a calibrated numerical experiment: assuming A and B are independently, jointly lognormally distributed with mean $A=100$, mean $B \in \{1, 10\}$, standard deviation of assets-in-place $\sigma_A \in \{10, 100\}$, standard deviation of project value $\sigma_B = 10$, required investment $I \in \{10, 100\}$ and slack S varied from 0 to 100 percent of I , the authors solve for the fixed point P' by iterating equations (1)–(2): start from $P' = S + A + B$; compute the resulting partition M, M' ; recompute P' from that partition, and repeat until convergence. This is a legitimate, if simple, numerical-methods exercise; the authors report (their footnote 15, drawing on Majluf's 1978 dissertation) that a unique fixed point of this kind was found in every lognormal and truncated-normal specification they tried, though they offer no general uniqueness proof.

Second, and distinct from the authors' own analysis, Section 3 of the paper marshals contemporaneous empirical working papers—*Dann and Mikkelson (1982)*, *Korwar (1982)* and *Asquith and Mullins (1983)*—as motivating evidence that seasoned equity announcements are met with significantly negative abnormal returns while debt announcements are not. **CRITIQUE** This evidence is not the authors' own empirical test of their model; it is external, and at the time of writing several of the cited studies (Asquith and Mullins in particular) were unpublished working papers, later appearing in print as *Asquith and Mullins (1986)*. The paper should therefore be read as a theoretical account constructed to explain a pattern that other researchers were independently documenting, rather than as a paper that tests its own model against data it collects.

5. Results

5.1 Equilibrium Properties

PAPER The model delivers a small set of clean, provable properties. The ex ante expected loss from underinvestment is $L = F(M) \cdot B(M)$, the probability of landing in the "do nothing" region multiplied by the average project NPV forgone there. $L = 0$ whenever $S \geq I$: sufficient slack fully restores first-

best investment. When $S < I$, L is increasing in the required issue $E = I - S$, so larger investment needs or thinner slack both raise the expected loss. The authors also prove (Section 2, "the impact of stock issues on stock price") that $P' < P$ whenever the probability of issuing is strictly between zero and one: the decision to issue conveys strictly negative information whenever it is not a foregone conclusion, because $A(M)$, the expected value of assets conditional on not issuing, must exceed $P' - S$ by the very logic of the partition.

5.2 The Numerical Simulation (Table 1)

Table 1. Illustrative ex ante loss in firm value (L , as a percentage of average project NPV $\bar{B}$), selected boundary cases

Slack (S , % of I)	$L/\bar{B}$, low σ_A	$L/\bar{B}$, high σ_A	Probability of issue $F(M')$
0 (no slack)	98–100%	98–100%	under 2%
50%	intermediate, falling in σ_A	intermediate, higher than adjacent column	rising with slack
90%	near 0–20%	still markedly above the low- σ_A column	rising sharply
100% ($S \geq I$)	0%	0%	100%

Source: Myers and Majluf (1984), Table 1 (underlying figures from Majluf, 1978, Tables 4 and 6). Only the fully legible boundary rows ($S = 0$ and $S = 100$ percent of I) are reproduced as point estimates from the scanned source consulted for this review; the intermediate rows ($S = 50, 90$ percent) are reported qualitatively because the digitised copy of the working paper is not fully legible at those cells. The direction of every comparative static reported here—loss falling in slack, falling in project NPV relative to investment size, and falling as asset-value uncertainty σ_A declines—is stated explicitly in the authors' own discussion of the table and is not in question.

EVIDENCE Three comparative statics survive clearly in the source text regardless of the illegible cells: (a) increasing slack monotonically reduces $L/\bar{B}$ and raises the probability of issuing, collapsing to zero loss once $S \geq I$; (b) increasing the project's average NPV relative to the investment required, B/I , reduces $L/\bar{B}$, because a sufficiently valuable project is worth issuing for even at unfavourable terms; (c) reducing the standard deviation of assets-in-place, σ_A , reduces the loss, consistent with the earlier proof that $L = 0$ exactly when $\sigma_A = 0$. No uniform effect of the standard deviation or correlation of B was found by the authors.

5.3 External Empirical Evidence Cited

SOURCED The magnitude the model predicts qualitatively—a negative price reaction to equity issues and a muted or absent reaction to debt—was subsequently confirmed with more data than the authors had in 1984. The published version of the study the authors cite as a working paper, *Asquith and Mullins (1986)*, reports an average abnormal return of approximately -2.7 percent on the announcement day for industrial seasoned equity offerings and a much smaller, though still negative, -0.9 percent for utilities; contemporaneous surveys such as *Smith (1986)* report two-day announcement windows of roughly -3.1 percent for industrials against -0.8 percent for utilities, and materially smaller or statistically insignificant effects for straight debt issues. **INFERENCE** Because industrial firms typically carry more intangible, harder-to-value assets than regulated utilities, the

wider gap for industrials is consistent with the model's prediction that the announcement discount should scale with the market's uncertainty about firm value, σ_A , though this cross-sectional comparison was not run as a formal test by the authors themselves.

6. Robustness and Additional Tests

Table 2. Robustness of the core result to the paper's own extensions

Concern	Extension examined	Result	Does it resolve the concern?
Issue costs might change the sign of the price effect	Add a fixed flotation cost T to the required gross issue	Firm may now issue projects with $0 < b < T$, a genuine negative-NPV-after-cost investment; the strict inequality $P' < P$ is no longer provable in general	Partially. The core qualitative story survives but the clean price-fall proof weakens; the authors flag this themselves and treat issue costs as second-order
The result might depend on financing being pre-announced rather than chosen with knowledge of a, b	Compare a pre-committed debt-or-equity policy to a policy chosen at $t=0$ after managers learn the truth	Discretionary choice strengthens rather than weakens the result: the firm never voluntarily issues equity once it knows a, b , because any equity price acceptable to the firm signals a sure loss to buyers	Yes, this robustness check runs in the theory's favour
The underinvestment might be an artefact of a static, one-shot model	Permanent versus temporary information asymmetry; building slack ahead of a future, still-unknown investment need	The same issue–invest logic reappears one period earlier, now applied to the decision to raise slack itself; the authors do not derive a fully optimal dynamic issue policy	No. The extension is argued informally and the authors explicitly decline to solve the dynamic problem
The result should have implications for mergers, not just security issues	A cash-rich acquirer bidding for a cash-poor target with private information	Negotiated mergers financed by information-revealing negotiation are shown to unravel in the baseline case; only unsolicited cash tender offers, or partial information disclosure, restore feasible deals	Consistent with, but not independently tested against, the empirical pattern that acquisitions are typically buyer-initiated

CRITIQUE None of these four checks is a robustness test in the modern empirical sense of re-running an estimation under alternative specifications or samples; they are analytical extensions of the same closed-form model. This is appropriate for a theoretical paper, but it means the model's robustness to its behavioural assumption—passive, non-rebalancing old shareholders—is never tested empirically within the paper itself, only asserted to be the necessary condition for the whole analysis (Section 3.4 above).

7. Critical Evaluation

7.1 What the Paper Actually Establishes

ESTABLISHED

Given passive old shareholders, rational investors and a two-asset joint claim, a fully rational equilibrium exists in which the firm sometimes declines a positive-NPV project rather than issue equity, and the decision to issue equity is unconditionally bad news whenever it is not certain. The comparative statics—loss decreasing in slack and in project NPV, increasing in asset-value uncertainty—are internally proved, not merely simulated.

SUPPORTED BUT CONDITIONAL

The ranking of debt over equity as an external financing source holds specifically when the information asymmetry concerns the level of firm value rather than its risk; the authors themselves note that *Giammarino and Neave (1982)*, who model asymmetric information about risk with the mean known, get the opposite ranking. The pecking order is therefore a conditional prediction about the type of information asymmetry a firm faces, not a universal law.

SUGGESTIVE

The paper's account of why negative announcement returns to equity issues obtain in practice is consistent with the empirical patterns cited, but consistency is not identification: *Miller and Rock (1982)*'s dividend-signalling model predicts the same sign of announcement effect for a different informational reason (unexpectedly high external financing reveals unexpectedly low current cash flow), and the 1984 paper does not construct a test that discriminates between the two mechanisms.

NOT ESTABLISHED

The paper does not establish that real-world managers behave as though maximising the interests of a class of shareholders assumed passive; that assumption is adopted because it "leads at this stage of the empirical race," in the authors' own words, not because it is independently derived or tested. It also does not establish the magnitude of underinvestment for any actual economy, sector or time period—the numerical results in Table 1 are calibration exercises under assumed lognormal distributions, not estimates from data.

7.2 Strengths

PAPER The authors themselves emphasise the model's ability to explain, in one internally consistent framework, four separate stylised facts: the preference for internal financing, the preference for debt over equity, the negative announcement return to equity issues, and the muted reaction to debt issues. **CRITIQUE** Independently of the authors' own framing, three further strengths stand out. First, the model is genuinely closed-form at its core (equations 1–2), which is

unusual for an asymmetric-information model of this period and is what allowed the pecking order prediction to be stated as a clean ranking rather than a numerical curiosity. Second, the authors are unusually candid about the fragility of their central behavioural assumption, walking the reader through exactly what happens to the result under active, rebalancing shareholders (Table 2) rather than burying that qualification in a footnote. Third, the debt-dominates-equity proof (Section 3.5) is a genuine theorem, not an assertion, resting on an explicit and checkable option-pricing inequality ($|\Delta D| < |\Delta E|$).

7.3 Weaknesses

WEAKNESS 1 — THE PASSIVE-SHAREHOLDER ASSUMPTION IS LOAD-BEARING AND UNMOTIVATED

Problem: the entire non-neutrality result depends on old shareholders being unable or unwilling to rebalance their portfolios in response to the firm's decision (Section 3.4). **Why it matters:** this is precisely the assumption Modigliani and Miller's logic would ordinarily rule out under frictionless markets, so the paper's novelty rests almost entirely on an assumption about investor behaviour rather than a market friction with independent empirical support. **Potential bias:** if real shareholders rebalance more than the model assumes—for instance, institutional holders with the scale to absorb pro-rata rights issues—the predicted underinvestment and price effects should be smaller for widely-held, institutionally-owned firms. **How it could be tested:** cross-sectionally, by checking whether announcement-period price declines and rights-issue take-up are smaller for firms with higher institutional ownership or lower shareholder-base dispersion, a test the original paper does not run.

WEAKNESS 2 — OBSERVATIONAL EQUIVALENCE WITH DIVIDEND-SIGNALLING AND OTHER BAD-NEWS MODELS

Problem: the model's headline empirical prediction (negative price reaction to equity announcements) is shared by at least one contemporaneous alternative mechanism, Miller and Rock's (1982) cash-flow-signalling model. **Why it matters:** without an independent test that varies the two mechanisms' distinguishing predictions—for example, whether the price reaction depends on issue size and pre-issue uncertainty as the pecking-order model predicts, versus on the deviation of financing from expected free cash flow as the dividend-signalling model predicts—confirmation of "a negative announcement return" does not discriminate between explanations. **Potential bias:** without discrimination, later researchers risk over-attributing all negative security-issue announcement effects to information asymmetry about firm value specifically. **How it could be tested:** interact announcement returns with proxies for information asymmetry (analyst dispersion, bid-ask spreads) versus proxies for cash-flow surprise, and check which proxy carries the explanatory power—a design later pursued in the capital-structure literature (see Section 9).

WEAKNESS 3 — THE NUMERICAL EXERCISE IS A CALIBRATION, NOT AN ESTIMATION

Problem: Table 1's loss estimates depend on assumed lognormal parameters for A and B that are illustrative rather than fitted to any observed distribution of firm values or project NPVs. Why it matters: the economic significance of the mechanism—how large the underinvestment problem actually is for real firms—cannot be read off this paper; the reader can only be confident of the sign and direction of the comparative statics, not their magnitude in practice. Potential bias: none directionally, but readers who cite the paper's numerical loss estimates as though they were empirical magnitudes would be misusing a calibration exercise. How it could be tested: structural estimation using firm-level distributions of forecast dispersion and project returns, an idea developed as a research opportunity in Section 11.

Of these three, the passive-shareholder assumption is the single most important weakness, because it is the assumption on which every other result in the paper depends, and it is adopted on grounds of tractability and consistency with the empirical announcement-return evidence rather than derived from a deeper primitive about trading costs, coordination problems among dispersed shareholders, or institutional constraints.

8. Contradictions and Edge Cases

INFERENCE The model's own internal logic identifies several conditions under which its central result disappears or reverses, and it is instructive to work through them explicitly.

WHEN DOES THE MECHANISM VANISH?

Three special cases collapse the underinvestment result to zero, each instructive about what is doing the work. If $\sigma_A = 0$ —the market already knows the value of assets in place—the firm always issues whenever $b \geq 0$, and $L = 0$ exactly (Section 2 of the paper, the "special cases" discussion). If $S \geq I$, slack alone restores first-best investment regardless of the asymmetry about assets in place. If old shareholders can be compelled to hold a pro-rata share of any new issue ($\alpha = 1$ in Table 2's notation), the Modigliani–Miller irrelevance result is restored even with full asymmetric information. Each of these is a genuine boundary of the theory rather than an assumption smuggled in, and each maps onto an observable firm characteristic—transparency, cash reserves, and shareholder concentration, respectively—that an empiricist could use to test the model out of sample.

WHAT HAPPENS AT THE EXTREME WHERE THE FIRM HAS NO INVESTMENT OPPORTUNITY?

PAPER When $b = 0$ in every state, the model breaks down in an economically informative way: the authors show that the only sustainable equilibrium has the firm issue stock only when assets in place are at their lower bound, a_{min} , and never otherwise—this is the classic Akerlof market-shutdown outcome, reproduced here as a special case of a more general model rather than assumed at the outset. **INFERENCE** This edge case is a useful diagnostic: any extension of the model

that fails to nest full market breakdown when the "good news" side of the joint claim vanishes should be treated with suspicion, because it would mean the extension has quietly relaxed the informational structure that generates the core lemons logic.

COULD A DIFFERENT MECHANISM GENERATE THE SAME ANNOUNCEMENT-RETURN PATTERN?

Yes, and the authors implicitly concede this by discussing Miller and Rock's (1982) alternative dividend-signalling channel. A negative announcement return to a security issue is also consistent with pure cash-flow signalling, with a resolution of agency costs of free cash flow (the free-cash-flow theory of *Jensen, 1986*, would in fact predict a positive reaction to a reduction in managerial discretion over cash in some settings, complicating the picture further), or with a change in expected systematic risk following the issue. The 1984 paper's evidence base, drawn from three then-unpublished working papers, cannot by itself adjudicate between these competing accounts.

DOES THE CONCLUSION SURVIVE OUTSIDE THE ORIGINAL TWO-STATE, ONE-SHOT SETTING?

EXTENSION The authors' own dynamic discussion (Section 4 of the paper) shows that with permanent information asymmetry the same partition logic reapplies one period earlier to the decision to build slack itself, so the qualitative conclusion is robust to at least one natural relaxation of the one-shot timing. What is not shown, and what the authors explicitly decline to solve, is whether a fully dynamic, repeated-game version of the model—in which a firm can build a reputation for using slack conservatively, or in which investors learn from a long history of issue decisions—preserves the same magnitude of underinvestment, or whether reputational forces erode it over time. This is precisely the kind of question a repeated-signalling extension (in the spirit of *Noe, 1988*) is built to answer, and it remains open within this paper.

9. Further Literature

Table 3. Literature connections — foundations and contemporaries

Paper	Shared idea	Relationship	Why it matters
Akerlof (1970), QJE 84, 488–500	Adverse selection shuts down trade in low-quality goods	Direct theoretical precursor; generalises the single-good lemons logic to a joint claim on two assets, one forfeited if the package is not sold	Establishes the paper's mechanism as a disciplined extension of a known result, not a new primitive
Ross (1977), Bell J. Econ. 8, 23–40; Leland and Pyle (1977), J. Finance 32, 371–387	Capital structure and ownership stakes as costly signals of firm quality	Contemporaneous signalling literature the paper departs from: no costly signal is needed here, only a rational break-even price conditional on the issue decision	Clarifies that the result needs only a partition equilibrium, not a Spence-style separating equilibrium
Jensen and Meckling (1976), JFE 3, 305–360	Capital structure choice under manager–shareholder–creditor conflicts	Complementary: the agency-cost framework explains why debt is costly at high leverage, capping the pecking order's debt preference	Shows why the pecking order is taught alongside, not instead of, static trade-off theory
Miller and Rock (1985), J. Finance 40, 1031–1051	Asymmetric information about current cash flow, signalled via financing and dividend surprises	Rival explanation for the same announcement–return evidence, with a different information structure	Identifies the empirical discrimination problem left unresolved by this paper (Section 7.3, Weakness 2)

Table 4. Literature connections — synthesis, extensions and later tests

Paper	Shared idea	Relationship	Why it matters
Myers (1984), J. Finance 39, 575–592	Static trade-off versus pecking order as competing descriptions of capital structure	Companion piece, same lead author and year, that names and popularises the "pecking order"	This paper is the formal micro-foundation; Myers (1984) is the synthesis that entered the curriculum
Asquith and Mullins (1986), JFE 15, 61–89; Krasker (1986), J. Finance 41, 93–105	Empirical and theoretical refinement of the announcement-return prediction	Asquith and Mullins supply the published large-sample evidence cited here only as a working paper; Krasker shows the decline should rise with relative issue size	Converts the qualitative sign prediction into a quantitative, testable cross-sectional one
Narayanan (1988), JFQA 23, 39–51; Noe (1988), RFS 1, 331–355; Lucas and McDonald (1990), J. Finance 45, 1019–1043	Refinements of the debt–equity signalling equilibrium and its dynamics	Extend the static partition equilibrium to richer security menus and to dynamic issue timing	Address the dynamic and security-choice questions this paper flags but does not solve
Shyam-Sunder and Myers (1999), JFE 51, 219–244; Frank and Goyal (2003), JFE 67, 217–248; Fama and French (2002), RFS 15, 1–33	Direct empirical tests of the pecking order against firm-level data	Shyam-Sunder and Myers find a good fit for large, mature firms; Frank and Goyal find a poor fit for small, high-growth firms; Fama and French find several departures from strict pecking-order predictions	The papers a reader should consult to judge whether the pecking order survives contact with broad data (Section 7 draws on their mixed verdict)

10. Research Gaps

PAPER-LEVEL GAP

The model never derives an optimal dynamic slack-building policy; it shows that the same partition logic reapplies to slack accumulation but explicitly stops short of solving for the optimal path of retention, dividends and issuance over time. Why it matters: without this, the model cannot generate a precise prediction for how quickly firms should rebuild slack after depleting it, only the qualitative direction. Existing limitation: the three-date structure was chosen for tractability, not because it captures how real financing decisions unfold repeatedly. Possible way forward: a fully dynamic contracting or real-options formulation of the slack-accumulation problem, in the spirit of the security-issue-timing literature that followed (Lucas and McDonald, 1990).

LITERATURE-LEVEL GAP

Decades after publication, large-sample tests of the pecking order (Shyam-Sunder and Myers, 1999; Frank and Goyal, 2003; Fama and French, 2002) disagree on whether it describes firm behaviour well, with the disagreement concentrated exactly among small, high-growth, informationally opaque firms that the theory was designed to describe. Why it matters: the theory's best-fitting population (large, seasoned firms with modest information asymmetry) is also the population in which the underlying mechanism should matter least, while its worst-fitting population is where the mechanism should bind hardest—an unresolved tension the empirical literature has not fully reconciled. Existing limitation: most large-sample tests proxy financing behaviour with aggregate leverage or deficit-financing regressions rather than directly measuring the ex ante information asymmetry the theory requires. Possible way forward: tests that directly measure information asymmetry (analyst dispersion, bid-ask spreads, earnings-surprise volatility) rather than inferring it from firm size or age as a proxy.

METHODOLOGICAL GAP

No study known to this reviewer has structurally estimated the model's own parameters—the joint distribution of A and B as perceived by the market—from real security-price and analyst-forecast data to recover an estimate of L , the ex ante value destroyed by information asymmetry, for actual firms. Why it matters: without this, the paper's numerical Table 1 remains an illustration of the mechanism's qualitative shape rather than a measurable economic magnitude. Existing limitation: the identification of A and B separately from observed prices is itself a hard inverse problem, since only their sum is typically priced before an issue is announced. Possible way forward: use options-implied or analyst-dispersion data around planned capital-expenditure announcements to separately proxy uncertainty about existing assets versus new projects, described further as Research Idea 5 below.

11. Research Opportunities

Idea 1 — Updating the Announcement-Return Test with Modern Information-Asymmetry Proxies

RESEARCH QUESTION: Has the magnitude of the negative announcement return to seasoned equity offerings shrunk as public disclosure, analyst coverage and real-time financial reporting have expanded since the 1980s and 1990s samples used in the founding studies?

MOTIVATION: The model predicts the announcement discount should scale with σ_A , the market's uncertainty about existing assets; if disclosure regimes (Regulation FD, mandatory XBRL filing, expanded analyst coverage) have genuinely reduced that uncertainty, the theory predicts a measurable decline in the average announcement discount over time.

HYPOTHESIS: The average announcement-period abnormal return to industrial SEOs has become significantly less negative, and the cross-sectional relationship between announcement returns and information-asymmetry proxies has weakened, in more recent decades relative to the 1980s baseline.

ECONOMIC MECHANISM: Improved public information narrows the dispersion of investor beliefs about assets in place, shrinking the region in which an issue signals overvaluation and thereby shrinking the required price concession.

LITERATURE CONNECTION: Extends Asquith and Mullins (1986) and Krasker (1986) forward in time; speaks directly to the Frank and Goyal (2003) finding that the pecking order fits informationally opaque firms poorly.

DATA: SDC Platinum or Refinitiv seasoned equity offering announcements, CRSP/Compustat returns and characteristics, I/B/E/S analyst dispersion, 1980–2025.

VARIABLES: Three-day cumulative abnormal return around the SEO announcement; analyst forecast dispersion, bid-ask spread, and 10-K/10-Q filing lag as information-asymmetry proxies; issue size relative to market capitalisation.

METHOD: Event-study estimation of abnormal returns by sub-period, followed by a panel regression of announcement returns on information-asymmetry proxies interacted with a post-Regulation-FD indicator.

IDENTIFICATION STRATEGY: Regulation FD (2000) and the expansion of mandatory XBRL filing (phased in 2009–2011) provide plausibly exogenous shifts in the public information environment usable as a difference-in-differences design around SEO announcements.

EXPECTED CONTRIBUTION: A direct, updated test of the theory's central comparative static using forty additional years of data and much richer information-asymmetry proxies than were available to Asquith and Mullins.

MAIN RISK: Confounding disclosure-regime effects with secular changes in the composition of issuing firms (more technology and biotech issuers with intangible assets, which could push the raw discount in the opposite direction).

Idea 2 — Textual Disclosure Complexity as a Direct Proxy for σ_A

RESEARCH QUESTION: Does the readability and information density of a firm's pre-issue disclosures (10-K filings) predict the size of its SEO announcement discount better than conventional size or age proxies for information asymmetry?

MOTIVATION: The model's key state variable, σ_A , is the market's uncertainty about assets already in place; textual disclosure complexity is a more direct proxy for that uncertainty than firm size, which conflates many other effects.

HYPOTHESIS: Firms with less readable, less specific 10-K disclosures in the year before an SEO experience significantly larger announcement-period price declines, controlling for size, leverage and issue size.

ECONOMIC MECHANISM: Harder-to-parse disclosure raises the market's residual uncertainty about the true value of assets in place, widening the region $\mathcal{M}'$ in which an issue pools good and bad realisations of a , and therefore deepening the required price concession.

LITERATURE CONNECTION: Builds a bridge between this paper's theoretical uncertainty parameter and the accounting and textual-analysis literature on disclosure readability that developed from the mid-2000s onward.

DATA: SEC EDGAR full-text 10-K filings, SDC SEO announcements, CRSP returns.

VARIABLES: Fog index or comparable readability score, filing length, boilerplate ratio, announcement CAR.

METHOD: Cross-sectional regression of announcement CAR on lagged disclosure-complexity measures, with industry and year fixed effects.

IDENTIFICATION STRATEGY: Within-firm changes in disclosure complexity around auditor or CFO turnover as a source of variation plausibly unrelated to the investment opportunity itself.

EXPECTED CONTRIBUTION: A more theory-consistent, continuously measured proxy for the paper's central uncertainty parameter than has previously been used in the empirical pecking-order literature.

MAIN RISK: Reverse causality: firms anticipating a difficult issue might deliberately simplify disclosure beforehand, biasing the readability measure.

Idea 3 — Cross-Country Disclosure-Regime Test

RESEARCH QUESTION: Do countries with stronger mandatory-disclosure regimes exhibit systematically smaller SEO announcement discounts, consistent with the model's prediction that the discount scales with investor uncertainty about firm value?

MOTIVATION: Cross-country variation in disclosure regulation (IFRS adoption dates, mandatory audit requirements, insider-trading enforcement intensity) offers plausibly exogenous variation in σ_A that a single-country study cannot exploit.

HYPOTHESIS: The SEO announcement discount is smaller, and less sensitive to firm-level opacity proxies, in country-years with stronger mandatory disclosure and enforcement.

ECONOMIC MECHANISM: Stronger disclosure regimes narrow investors' ex ante uncertainty about assets in place economy-wide, shrinking the pooling region M' for all issuing firms in that jurisdiction.

LITERATURE CONNECTION: Extends the pecking-order testing literature (Frank and Goyal, 2003) into an international, regulation-driven identification design rather than a purely time-series or single-country cross-section.

DATA: Worldscope/Refinitiv international SEO announcements, disclosure and enforcement indices from the law-and-finance literature, staggered IFRS adoption dates by country.

VARIABLES: Announcement CAR, country-level disclosure index, IFRS adoption indicator, firm-level opacity proxies.

METHOD: Staggered difference-in-differences around IFRS adoption dates, with firm and country-year fixed effects.

IDENTIFICATION STRATEGY: Staggered mandatory IFRS adoption across countries between the late 1990s and 2010s as a quasi-natural experiment in disclosure quality.

EXPECTED CONTRIBUTION: The first cross-country causal test of the model's core comparative static using a regulation-driven, rather than purely cross-sectional, source of identification.

MAIN RISK: IFRS adoption is correlated with many concurrent capital-market reforms, making it hard to isolate the disclosure channel specifically.

Idea 4 — Verifiable Treasuries and the Value of Slack in On-Chain Organisations

RESEARCH QUESTION: Do organisations with continuously, publicly verifiable balance sheets (on-chain treasuries of decentralised autonomous organisations) exhibit a smaller wedge between internal and external financing costs than comparable conventional firms?

MOTIVATION: The model's mechanism depends entirely on the market being unable to verify a , the value of assets in place, at the moment of financing; a treasury that is verifiable in real time by construction offers a rare setting in which σ_A is close to zero by design.

HYPOTHESIS: Token or equity issuances by organisations with fully transparent, on-chain treasuries show a significantly smaller announcement discount than issuances by organisations with opaque or unverifiable reserves, controlling for size and issue characteristics.

ECONOMIC MECHANISM: Verifiable reserves collapse the region over which an issue can pool high- and low-value realisations of assets in place, directly testing the paper's own comparative-static prediction that L approaches zero as σ_A approaches zero.

LITERATURE CONNECTION: A genuinely new empirical laboratory for the paper's central prediction, situated at the intersection of this paper's asymmetric-information mechanism and the emerging financial-technology literature on transparent treasury management.

DATA: On-chain treasury data from DAO governance platforms, token issuance announcements and secondary-market price reactions.

VARIABLES: Announcement-window abnormal token return, treasury verifiability indicator, treasury size relative to raise size.

METHOD: Event-study comparison of announcement returns across verifiable- versus opaque-treasury issuers, matched on raise size and sector.

IDENTIFICATION STRATEGY: Cross-sectional comparison exploiting the discrete, technologically determined difference in verifiability rather than a continuous proxy.

EXPECTED CONTRIBUTION: A clean, near-laboratory test of the theory's limiting case that is unavailable in conventional equity markets.

MAIN RISK: Small, young, and highly volatile sample; token markets have idiosyncratic frictions (regulatory uncertainty, thin liquidity) that could dominate any information-asymmetry effect.

Idea 5 — Structural Estimation of the Value of Financial Slack

RESEARCH QUESTION: What is the ex ante dollar value of financial slack, L , implied by the Myers–Majluf model when its parameters are estimated from real firm-level data rather than assumed, and does observed corporate cash-holding behaviour track the model-implied optimal slack?

MOTIVATION: The paper's own numerical results are illustrative calibrations (Section 5.2); no study has structurally recovered the model's parameters from data to produce a firm-specific estimate of the value of slack, which would let the theory be judged on economic magnitude rather than sign alone.

HYPOTHESIS: Firms hold financial slack in a magnitude that co-moves with the model-implied optimal slack derived from their own estimated σ_A and investment-opportunity distribution.

ECONOMIC MECHANISM: Same as the original paper; the contribution here is purely estimation, mapping analyst-forecast dispersion and options-implied volatility into estimates of A and B 's joint distribution and then solving the fixed point in equations (1)–(2) firm by firm.

LITERATURE CONNECTION: Directly operationalises the "methodological gap" identified in Section 10 and complements the reduced-form pecking-order tests of Shyam-Sunder and Myers (1999) and Frank and Goyal (2003) with a structural counterpart.

DATA: I/B/E/S analyst forecast dispersion, OptionMetrics implied volatility, Compustat cash holdings and capital-expenditure plans.

VARIABLES: Estimated σ_A and σ_B , observed cash-to-assets ratio, capital-expenditure realisation rates.

METHOD: Simulated method of moments, fitting the model's predicted cash-holding and issuance frequency to observed firm-level moments.

IDENTIFICATION STRATEGY: Uses the model's own equilibrium conditions as the estimating equations, identified off cross-sectional variation in forecast dispersion and investment-opportunity volatility.

EXPECTED CONTRIBUTION: The first structural, rather than calibrated or reduced-form, estimate of the economic magnitude of the Myers–Majluf financing friction.

MAIN RISK: Separately identifying uncertainty about assets in place from uncertainty about the new project is econometrically difficult, since both typically move the same observed price.

Ranking of Research Ideas

Ranked by novelty, feasibility, data availability, methodological quality and economic importance, the top three are Idea 1, Idea 5 and Idea 3. Idea 1 ranks highest on feasibility and data availability: it uses well-established event-study methodology and widely available databases, and offers the most direct, lowest-risk update of the paper's central empirical claim using forty additional years of evidence. Idea 5 ranks highest on novelty and methodological ambition: no existing paper has structurally estimated the model's own parameters, and doing so would convert Table 1's illustrative calibration into a genuine economic magnitude, though execution risk around separately identifying σ_A and σ_B is real. Idea 3 offers the strongest causal identification of the three, exploiting staggered IFRS adoption as a quasi-natural experiment, at the cost of requiring an international dataset that is more expensive to assemble than a single-country study. Ideas 2 and 4

are promising but ranked lower: Idea 2 faces a plausible reverse-causality confound that would need to be resolved before the results could be read as causal, and Idea 4's sample of transparent-treasury organisations is currently too small and too idiosyncratic to support strong inference.

12. Learning Framework

Prerequisite Concepts

Net present value and the Modigliani–Miller capital-structure irrelevance theorems; the concept of a rational-expectations equilibrium in which prices are set to be consistent with the information revealed by agents' own equilibrium actions; the basic logic of Akerlof's market for lemons; elementary option-pricing intuition (the relationship between the riskiness of a claim and the sensitivity of its value to new information, used in the debt-versus-equity proof).

Core Concepts Learned

A partition equilibrium, in which a continuous state space is divided into an "act" region and a "do not act" region by a zero-profit break-even price, as an alternative to the discrete-type separating and pooling equilibria of the standard Spence signalling model; the concept of financial slack as a real option to invest without incurring an information-driven price discount; the pecking order as an equilibrium ranking of financing sources rather than a behavioural rule of thumb.

Important Equations

The issue condition, $(E/P')(S+a) < E+b$ (equation 1); the rational-expectations break-even price, $P' = S + A(M') + B(M')$ (equation 2); the debt-versus-equity issue condition in terms of the ex post capital gain to new security holders, $b > \Delta E$ or $b > \Delta D$ (equation 3), together with the inequality $|\Delta D| < |\Delta E|$ that drives the pecking-order ranking.

Methods Learned

Solving a signalling-type equilibrium by fixed-point iteration on a break-even price rather than by backward induction over discrete types; conducting comparative statics on an ex ante expected-loss functional; using a numerical simulation to illustrate, rather than to test, a closed-form theoretical result.

Questions to Test Understanding

Comprehension. Explain in your own words why old shareholders in the model's two-state example are better off withholding the equity issue in the high-value state, even though this means forgoing a positive-NPV project.

Analysis. Explain precisely why the underinvestment result depends on old shareholders being passive, and show, using Table 2's logic, what happens to the issue-invest decision if old shareholders are instead assumed to hold a pro-rata share of every new issue.

Research. Design an empirical test that could distinguish the Myers-Majluf mechanism (information asymmetry about firm value) from the Miller-Rock mechanism (information asymmetry about current cash flow) using announcement-return data; what variable would you need that neither paper's original evidence base contained?

Papers to Read Next

Myers (1984), "The Capital Structure Puzzle," for the accessible synthesis naming the pecking order; *Shyam-Sunder and Myers (1999)* and *Frank and Goyal (2003)*, for the large-sample empirical verdict on whether the pecking order fits observed financing behaviour; *Krasker (1986)*, for the direct extension linking announcement returns to relative issue size.

13. Conclusion

ONE-SENTENCE THESIS

When managers know more than investors about the value of a firm's assets and its growth opportunities, a rational market will always read a stock issue as bad news, and this alone—absent any tax, distress or transaction cost—can lead value-maximising firms to forgo genuinely valuable investments and to rank financing sources from internal funds, to debt, to equity as a last resort.

THREE MOST IMPORTANT INSIGHTS

First, financing non-neutrality can arise purely from information frictions, with no need for taxes or bankruptcy costs. Second, financial slack is not idle capital but a real option whose value is precisely the underinvestment loss it prevents. Third, the debt-over-equity preference is conditional on the type of asymmetric information (value versus risk), not a universal ranking, a nuance frequently lost in simplified expositions of the pecking order.

STRONGEST EVIDENCE

The proof that $P' < P$ whenever the probability of issuing is strictly between zero and one, because $A(M) > P' - S$ follows directly from the structure of the partition itself; this is a genuine theorem, not a simulated tendency, and it is the result that later empirical work (Asquith and Mullins, 1986; Smith, 1986) went on to confirm at approximately the magnitude the theory implies for informationally opaque, industrial issuers.

BIGGEST WEAKNESS

The passive-shareholder assumption is unmotivated by any deeper primitive and is precisely the assumption a frictionless-markets baseline would rule out; the entire non-neutrality result rests on it, and the paper does not test whether real shareholders behave this way.

MOST IMPORTANT RESEARCH GAP

No study has structurally estimated the model's own parameters from firm-level data to produce a measurable, rather than illustrative, economic magnitude for the value of financial slack, leaving decades of qualitative confirmation without a corresponding quantitative benchmark.

BEST RESEARCH OPPORTUNITY

Re-running the announcement-return test with modern, direct measures of information asymmetry across a period spanning major disclosure-regime changes (Research Idea 1) offers the best combination of feasibility, data availability and a clean link back to the paper's own central comparative static.

THREE PAPERS TO READ NEXT

Myers (1984), The Capital Structure Puzzle, Journal of Finance 39, 575–592; *Frank and Goyal (2003), Testing the Pecking Order Theory of Capital Structure*, Journal of Financial Economics 67, 217–248; *Krasker (1986), Stock Price Movements in Response to Stock Issues under Asymmetric Information*, Journal of Finance 41, 93–105.

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